RISHAV MONDAL

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SUMMARY

Machine Learning Engineer with 3+ years of experience building scalable machine learning solutions and predictive models using structured and unstructured data. Skilled in Python, statistical analysis, and deriving insights from raw datasets to support business goals. Proven track record in enhancing model performance, optimizing workflows, and enabling data-driven decision-making.

EDUCATION

Master of Science in Business Analytics, Clark University - *Worcester, MA* May 2025

Master of Science in Computer Science, University of Milan – *Milan, Italy* May 2023

EXPERIENCE

Data Engineer
Community Dreams Foundation - *Remote* **June 2025 - Present**

- Build scalable Azure pipelines using Azure Data Factory, Databricks, and Python, to reduce data ingestion time by 70%.
- Design SQL and NoSQL data models to support analytics and ML, with aim to streamline integration across 5+ systems.
- Develop batch and real-time workflows with automated testing and monitoring, projected to cut pipeline failures by 80%.
- Improve data accessibility and performance by implementing ML-ready architecture, enhancing decision-making speed and accuracy across teams.

Marketing Research Analyst
Clark University Campus Store – *Worcester, MA* **Feb 2025 - May 2025**

- Created a survey and analyzed 2,900+ responses and 473 transactions using Excel, Python, and NLP to uncover customer sentiment and behavior trends and KPIs, leading to a 23% increase in customer satisfaction.
- Uncovered a 97% online awareness gap and performed A/B testing and statistical analysis to validate an Instagram-first strategy, boosting digital engagement by 57%.
- Identified high-demand products through product mix analysis and drove targeted campaigns for student exclusive deals projecting a 15–20% increase in per-customer revenue.

Machine Learning Research Engineer
University of Milan – *Milan, Italy* **Feb 2022 - Sep 2024**

- Led the development of an Optical Music Recognition system processing 60GB+ of handwritten manuscript images, enabling automatic classification and digital restoration of historical scores.
- Developed scalable pipelines for symbol detection and fine-tuned deep learning models in PyTorch, achieving 88% accuracy and creating a benchmark, and outperforming baseline classifiers by 20%+.
- Engineered and implemented ETL workflows for 473K+ images, improving data handling and cutting manual workload by 30% and boosting scalability for future enhancements.
- Refined classification accuracy through EDA and class merging on 198K+ labeled images, improving annotation consistency and inter-annotator agreement by 17%.
- Published a paper in ACM on Optical Music Recognition (OMR) using deep learning; presented at AudioMostly2024 conference, showing approximately 20% accuracy improvement over prior benchmarks and earning industry recognition.

PROJECTS

Heart Disease Prediction **Dec 2024**

- Engineered scalable data pipelines using PySpark and SQL DataFrame APIs to analyze large-scale healthcare provider data and identify key risk drivers, enhancing operational risk intelligence and compliance reporting.
- Designed and deployed classification models, achieving up to 91% accuracy; to ensure reliable risk prediction.

Twitter Sentiment Analysis **May 2024**

- Collected and processed 1.6M tweets using the Twitter API; performed sentiment analysis with NLTK to extract public opinion trends, supporting real-time brand monitoring and audience engagement strategies.
- Evaluated and optimized models achieving 67% accuracy to extract actionable insights and support sentiment-driven decisions.

SKILLS

Data and AI Governance, Supervised & Unsupervised Learning, Feature Engineering, Model Development & Evaluation, Statistical Analysis, ELT/ETL Pipelines, Big Data Processing, Database Management, Monte Carlo Simulation, Data Science.

TECHNICAL SKILLS AND CERTIFICATIONS

Data Analytics Tools: Looker Studio, Tableau, MS Excel, MongoDB, Cassandra, Spark, Power Bi, SQL/NoSQL, SSIS.
Programming Languages and Cloud Platforms: Python, Java, MATLAB, R, AWS S3, Azure Databricks, KNIME, C++.
Machine Learning: Decision Trees, Natural Language Processing(NLP), Predictive Modeling, CUDA, Uncertainty Modeling.
Frameworks: scikit-learn, TensorFlow, Keras, StatsModel, Pandas, NumPy, Matplotlib, Seaborn, PySpark, Git, Snowflake.
Certification: Data Camp Power BI, Google GA4, Data Camp SQL, PCEP, Forage BCG Data Science.